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EXPERIMENT 14

Transformation using a Homography Matrix

AIM

To transform an image using a homography matrix estimated with cv2.findHomography().

PROGRAM

```
import cv2
import numpy as np

img = cv2.imread("image1.jpg")
rows, cols = img.shape[:2]

src_points = np.float32([[0, 0], [cols - 1, 0],
                         [0, rows - 1], [cols - 1, rows - 1]])
dst_points = np.float32([[0, 0], [cols - 1, 0],
                         [0, int(0.7 * rows)], [cols - 1, int(0.7 * rows)]])

M, status = cv2.findHomography(src_points, dst_points)
homography_img = cv2.warpPerspective(img, M, (cols, rows))

cv2.imwrite("homography_image.jpg", homography_img)
print("Homography matrix:\n", M)

cv2.imshow("Original", img)
cv2.imshow("Homography Transformed", homography_img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

INPUT

image1.jpg (input image)

OUTPUT

homography_image.jpg (the image is compressed to 70% of its height)

RESULT

Thus the given image was successfully transformed using a homography matrix.